House Prices in Ames, Iowa



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Big Data Management Systems and Tools: Group 7

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1. Background

House prices incorporate many factors. Over the years many features have impacted real estate pricing in different ways. This project presents a data set describing the sale of individual residential property in Ames, Iowa from 2006 to 2010. The data set contains 2919 observations and 79 explanatory variables involved in assessing home values.

The need for a model to predict the house price based on a variety of key features can be very important for people looking to buy a house. Looking at the correlation between certain variables can lead to some interesting insights. For individuals looking to become homeowners and real estate agents looking to assess the housing market, this information would be extremely beneficial in terms of determining fair value for a home.

By employing various data models (Linear Regression, Random Forest, Boosted Tree Regressor, Gradient Boosted Tree Regressor) against the available training set we aim to create a predictor for determining house prices in Anes, Iowa. The accuracy of our predictor will be heavily influenced by specific attributes included in our Kaggle dataset.

2. Objective

Our goal is to build a predictive model that would determine the price of houses in Ames, Iowa based on a variety of provided features. By employing various data science tools and forecasting techniques our team aims to have a model that can successfully predict and approximate the sale price of homes in Ames, Iowa. Additionally, our analysis will seek to understand the relationship between key variables and the sale price of homes in the region. By fulfilling our objective our team will aim to prove how this information can be incredibly valuable for individuals seeking to purchase property as they would already know a house's fair value price. Individuals that possess knowledge about a house's fair value price will be at an incredible advantage with

regards to making an informed and educated decision when investing in a home, avoiding potential manipulation and scams in the process.

Our goal to produce data models that can successfully forecast and predict the sale prices of house in Ames, Iowa, will be accomplished by following established methodology in data science regarding data analysis and model creation. Firstly, our team will set out to clean and prepare the Kaggle dataset we are utilizing to create our data models. This will be accomplished by substituting any missing values (NAs) in a data column by either using that attribute's median or mode or combining information from other related attributes. The following step will be to analyze the data and drop any attributes that are determined to not have sizable influence on the sale price of houses in the region.

Finally, our team will create multiple data models that employ both a training and validation set. The validation set will establish which data model has the highest level of accuracy in terms of predicting the sale price of homes. This will be beneficial for any individual looking utilize the best available data model when assessing the real estate market in the region. When creating these data models, a separate analysis will be conducted to determine which attributes had the highest level of influence on the sale price. This analysis will provide valuable insights in terms of determining which features of a home are the most coveted for those purchasing housing property in Ames, Iowa. The attributes we utilized have been included below along with their corresponding definitions.

Glossary

- **SalePrice -** the property's sale price in dollars. This is the target variable that you're trying to predict.
- **MSSubClass:** The building class
- **MSZoning:** The general zoning classification
- **LotFrontage:** Linear feet of street connected to property
- **LotArea:** Lot size in square feet
- **Street:** Type of road access
- **Alley:** Type of alley access
- **LotShape:** General shape of property
- **LandContour:** Flatness of the property
- **Utilities:** Type of utilities available
- **LotConfig:** Lot configuration
- **LandSlope:** Slope of property
- **Neighborhood:** Physical locations within Ames city limits
- **Condition1:** Proximity to main road or railroad
- **Condition2:** Proximity to main road or railroad (if a second is present)
- **BldgType:** Type of dwelling
- **HouseStyle:** Style of dwelling
- **OverallQual:** Overall material and finish quality
- **OverallCond:** Overall condition rating
- **YearBuilt:** Original construction date
- **YearRemodAdd:** Remodel date
- **RoofStyle:** Type of roof
- **RoofMatl:** Roof material
- **Exterior1st:** Exterior covering on house
- **Exterior2nd:** Exterior covering on house (if more than one material)
- **MasVnrType:** Masonry veneer type
- **MasVnrArea:** Masonry veneer area in square feet
- **ExterQual:** Exterior material quality
- **ExterCond:** Present condition of the material on the exterior
- **Foundation:** Type of foundation
- **BsmtQual:** Height of the basement
- **BsmtCond:** General condition of the basement
- **BsmtExposure:** Walkout or garden level basement walls
- **BsmtFinType1:** Quality of basement finished area
- **BsmtFinSF1:** Type 1 finished square feet
- **BsmtFinType2:** Quality of second finished area (if present)
- **BsmtFinSF2:** Type 2 finished square feet
- **BsmtUnfSF:** Unfinished square feet of basement area
- **TotalBsmtSF:** Total square feet of basement area
- **Heating:** Type of heating
- **HeatingQC:** Heating quality and condition
- **CentralAir:** Central air conditioning
- **Electrical:** Electrical system
- **1stFlrSF:** First Floor square feet
- **2ndFlrSF:** Second floor square feet
- **LowQualFinSF:** Low quality finished square feet (all floors)
- **GrLivArea:** Above grade (ground) living area square feet
- **BsmtFullBath:** Basement full bathrooms
- **BsmtHalfBath:** Basement half bathrooms
- **FullBath:** Full bathrooms above grade

- **HalfBath:** Half baths above grade
- **Bedroom:** Number of bedrooms above basement level
- **Kitchen:** Number of kitchens
- **KitchenQual:** Kitchen quality
- **TotRmsAbvGrd:** Total rooms above grade (does not include bathrooms)
- **Functional:** Home functionality rating
- **Fireplaces:** Number of fireplaces
- **FireplaceQu:** Fireplace quality
- **GarageType:** Garage location
- **GarageYrBlt:** Year garage was built
- **GarageFinish:** Interior finish of the garage
- **GarageCars:** Size of garage in car capacity
- **GarageArea:** Size of garage in square feet
- **GarageQual:** Garage quality
- **GarageCond:** Garage condition
- **PavedDrive:** Paved driveway
- **WoodDeckSF:** Wood deck area in square feet
- **OpenPorchSF:** Open porch area in square feet
- **EnclosedPorch:** Enclosed porch area in square feet
- **3SsnPorch:** Three season porch area in square feet
- **ScreenPorch:** Screen porch area in square feet
- **PoolArea:** Pool area in square feet
- **PoolQC:** Pool quality
- **Fence:** Fence quality
- **MiscFeature:** Miscellaneous feature not covered in other categories
- **MiscVal:** $Value of miscellaneous feature
- **MoSold:** Month Sold
- **YrSold:** Year Sold
- **SaleType:** Type of sale
- **SaleCondition:** Condition of sale

3. Data Overview

1. There are exactly 2919 entries of the non-cleaned house sale records in Ames, Iowa in the chosen dataset. The entries contain 79 attributes of each house.
2. The column names were examined to ensure that they did not contain any unusual characters or ambiguous names.
3. There were _ attributes within the dataset that were not missing any values while approximately _ attributes had at least one or more NAs in its data column. This would indicate that the dataset is mostly complete though certain attribute's data values could be filled using various methods.

4. Data Preparation

The dataset used for the project is open-sourced and obtained through Kaggle (https://www.kaggle.com/c/house-prices-advanced-regression-techniques/data). Upon visually inspecting the dataset we proceeded with cleaning the data by dropping any attributes with more than 30 percent missing values (NA/Null). Further inspection of the data led us to the conclusion that these attributes would not impact the sale price in any significant way. Our team also wanted to avoid overfitting or overtraining the data models that could lead to obtaining misleading results. As a result, we dropped numerous attributes outright, including 'PoolQC', 'MiscFeature' 'Alley', 'Fence', and 'FireplaceQu'. With the amount of missing data for these attributes it did not seem as though they would be particularly impactful when predicting the sale price of houses. Numeric attributes with missing values had their data filled and replaced with either that attribute's mode or median. Missing values for non-numeric attributes were filled in by referring to data from two related attributes (Eg. LotFrontage). An explanation of how we replaced missing data for specific attributes is included below:

Numeric Attributes

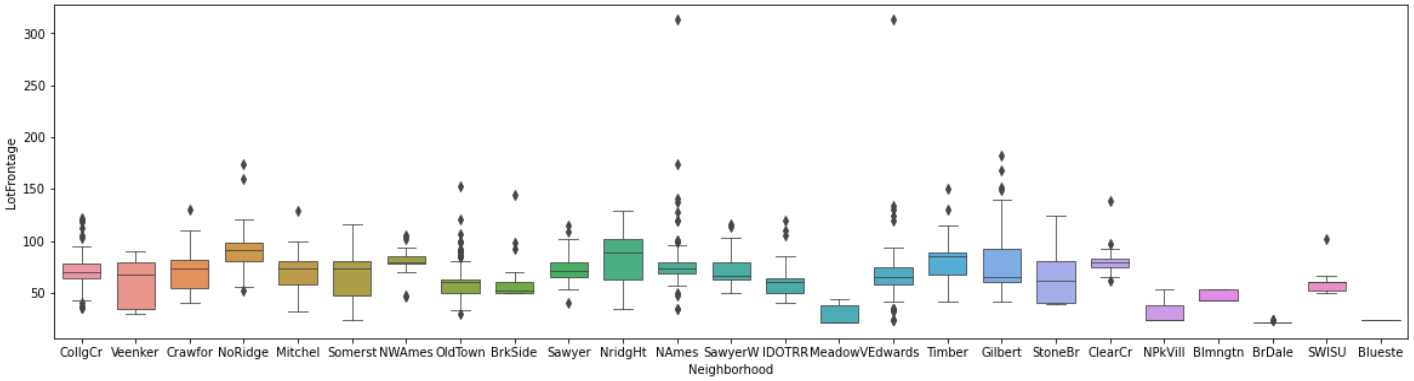
a) **Electrical:** This attribute is being filled with the most common type (mode).

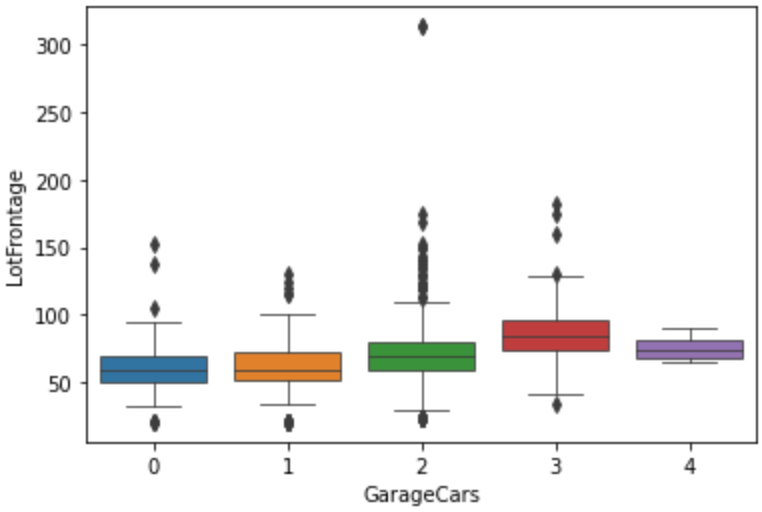
b) **Basement Attributes:** It is important to note that NA in bsmtQual, bsmtCond, bsmtExposure, bsmtFinType1, and bsmtFinType2, indicates that there is no basement, which can be proven by an indicator TotalBsmtSF. If TotalBsmtSF displays 0, it means that there is no basement within the home. To handle NAs/Null Values in the Basement Attributes, NAs will be replaced by 'No Basement' if TotalBsmtSF is 0. For the rest missing values, mode is used to replace the missing data.

c) **Garage attributes:** It is important to note that NA in Graebel, GarageFinish, GarageQual, & GarageCond indicates that there is no garage within the home, which can be proven by an indicator GarageArea. If the attribute GarageArea displays 0, it means there is no garage within the home. To handle NAs/Null Values in the Garage Attributes, NAs will be replaced by 'No Garage' if GarageArea is 0. For the rest of the missing values, mode is used to replace the missing data. Online resources also indicate that at best only 78% of US homes have garages/carports. In our case, just under 6% of homes do not have garages. If anything, it appears only 5.5% of homes in Ames, Iowa do not have garages. (https://www.energy.gov/eere/vehicles/fact-958-january-2-2017-sixty-three-percent-all-housing-units-have-garage-or-carport).

Non-Numeric Attributes

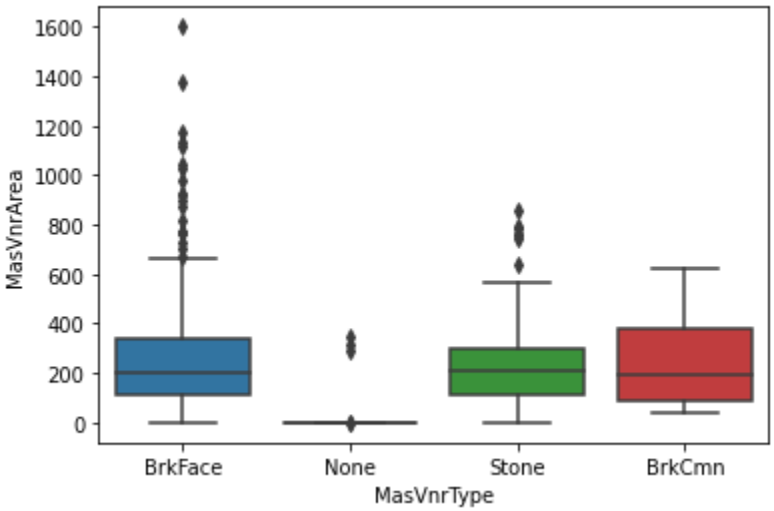
LotFrontage: The LotFrontage attribute is related to the Neighborhood and GarageCar attribute. This was proven by creating a box plot to analyze and ensure the validity of this relationship. Findings indicated that as the numbers of cars a home's garage could support increased, the lot frontage of that home would also rise. Hence grouping the data by Neighborhood and GarageCar should form distinct groups whose median can be used as a value to fill in NaNs for LotFrontage.





GarageYrBlt: Based on the observation of the raw data, it can be noticed that the missing value in this column is shown as NA. It is also worth noting that all NAs are associated with the 0 value in the GarageArea, which means there is no garage. So NAs here are being replaced by -1.

MasVnrArea: To fill the null values in MasVnrAreaTo, a boxplot was created to find if there is any relationship between MasVnrArea and MasVnrTyp.



From the plot above, the MasVnrArea for different types of Masonry Veneer is practically identical therefore there should not be any relationship between MasVnrType and MasVnrArea. If we examine the data in MasVnrArea, we notice that the value 0 implies that there is no Masonry veneer since MasVnrType value is None. For the rest of the NA values in MasVnrArea we simply replaced it with the median of that attribute.

5. Data Analysis

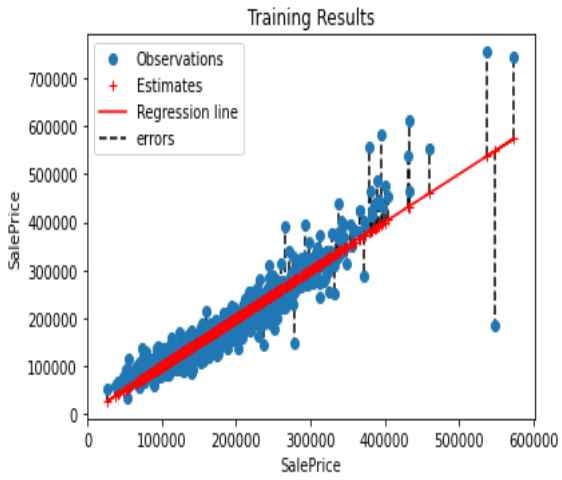
Upon successfully cleaning and prepping the dataset for house prices our team proceeded with data analysis. The final attributes list of attributes that would be included for data analysis would be the following: 'Id', 'MSZoning', 'Street', 'LotShape', 'LandContour', 'Utilities', 'LotConfig', 'LandSlope', 'Neighborhood', 'Condition1', 'Condition2', 'BldgType', 'HouseStyle', 'RoofStyle', 'RoofMatl', 'Exterior1st', 'Exterior2nd', 'MasVnrType', 'MasVnrArea', 'ExterQual', 'ExterCond', 'Foundation', 'BsmtQual', 'BsmtCond', 'BsmtExposure', 'BsmtFinType1','BsmtFinType2', 'Heating', 'HeatingQC', 'CentralAir', 'Electrical', 'KitchenQual', 'Functional', 'GarageType', 'GarageFinish', 'GarageQual', 'GarageCond', 'PavedDrive', 'SaleType', 'SaleCondition' and 'SalePrice'.

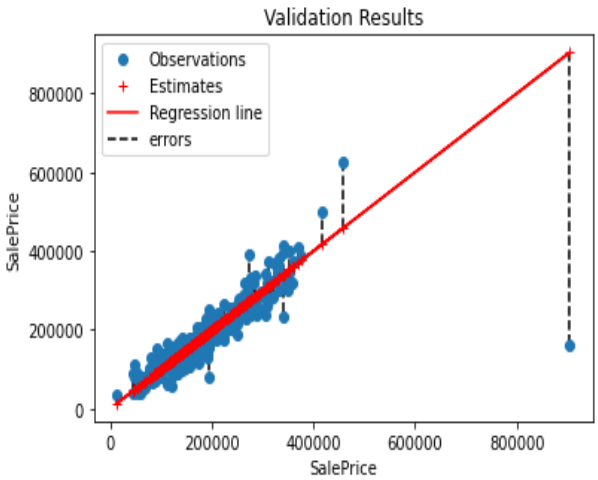
The data models we utilized to create out predictor for house prices in Ames, Iowa was Linear Regression, Random Forest, Boosted Tree Regressor and Gradient Boosted Tree Regressor. By running these data models against the available training set our team was able to determine the accuracy of our predictions with regards to house prices. Each model we employed contained their own set of advantages when determining house prices, though our goal upon further study of these models will be to determine which model is most suitable with regards to our analysis. The accuracy of our data models will be verified by comparing our training and validation set. Two key components used in this analysis will be **Root Mean Square Error** (RMSE) and **r^2**. RMSE will act as a standard way to measure the error of a model by measuring the differences between values predicted by the estimators in our model and the values observed while r^2 will measure the proportion of the variation in the dependent variable that is predictable from the independent variable. A lower RMSE and a higher r^2 within our validation set will indicate that the model has a higher level of accuracy when

predicting outcomes. Using these data science methodologies our team was able to determine which data model was the most precise with regards to predicting the value of a home by analyzing its relevant attributes. Details regarding each data model that our team utilized have been included below:

5.1: Linear Regression

Our group was to employ a linear regression model to determine whether there happened to be any positive relationship between the attributes we utilized for data analysis and house prices. Our first step was to import any dependencies for our linear regression model. Once complete our team created the linear regression model based on sales price and the relevant features. Upon doing so our team proceeded to fit the model and transform the validation set. Finally, we employed an estimator to determine the accuracy of the data against the training results versus the validation results. Our analysis concluded that though our estimator showed a high level of correlation and a lower root-mean square error within our training set (r^2 = 86.79%, RMSE = 28755.62), our validation results posed a different picture. Within our validation results, we had a low level of correlation (66.70%) and a very high RMSE (46268.30). This would indicate that this data model may not be the most suitable when predicting house prices in Ames, Iowa.

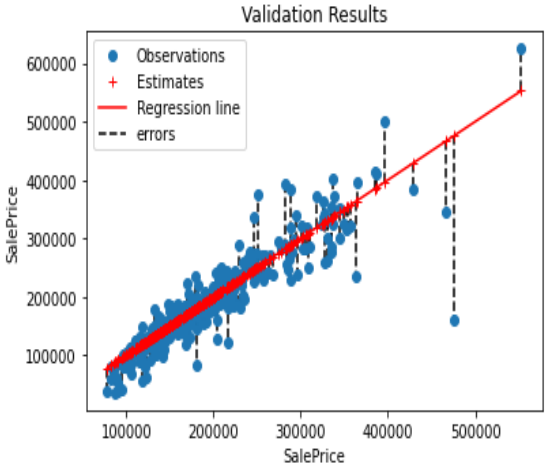




5.2: Random Forest

Our group was to employ a random forest model to determine whether there happened to be any positive relationship between the attributes we utilized for data analysis and house prices. Our first step was to import any dependencies for our linear regression model. Once complete our team created the linear regression model based on sales price and the relevant features. Upon doing so our team proceeded to fit the model and transform the validation set. Finally, we employed an estimator to determine the accuracy of the data against the training results versus the validation results. Our analysis concluded that our estimator showed a similar level of correlation and RMSE between the training set and validation set. Though our training set did perform better (r^2 = 97.66%, RMSE = $12099.61) with a higher level of correlation and a lower RMSE than our validation results ((r^2 = 86.17%, RMSE = $29823.68), it still performed well with regards to accurately predicting house prices in Ames, Iowa.

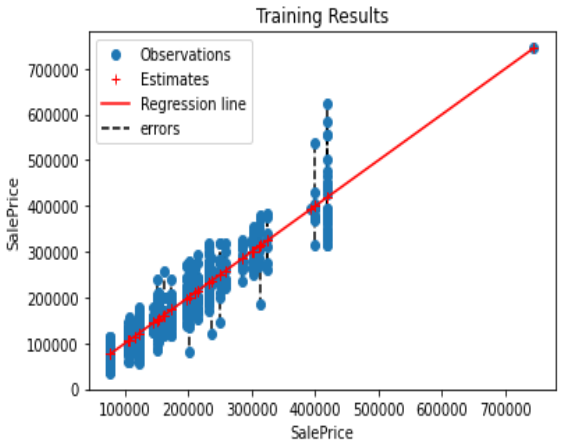


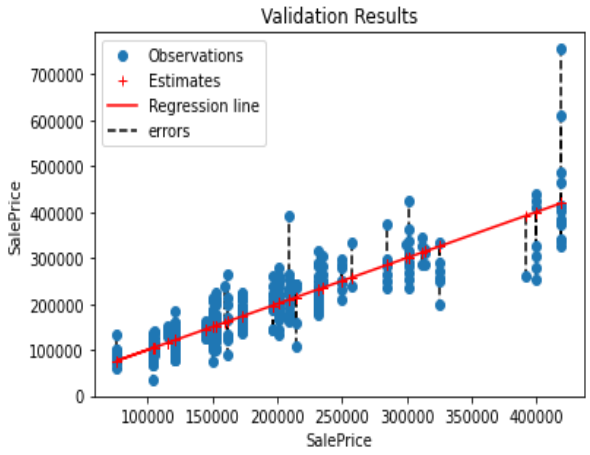


5.3: Decision Tree Regression

Our analysis concluded that our estimator showed a high level of correlation and a lower root-mean square error within our training set (r^2 = 86.20%, RMSE = 29242.25), our validation results posed a different picture. Within our validation results, we had a decent level of correlation (76.90%) and a higher RMSE (38899.67). This would indicate

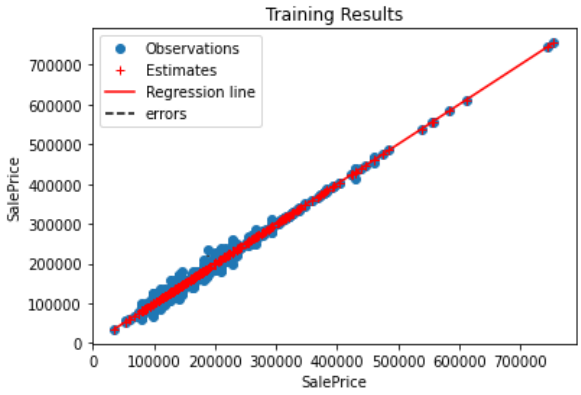
that this “Decision tree regression” may not be the most suitable when predicting house prices in Ames, Iowa due to the higher level of RMSE compared to other models.

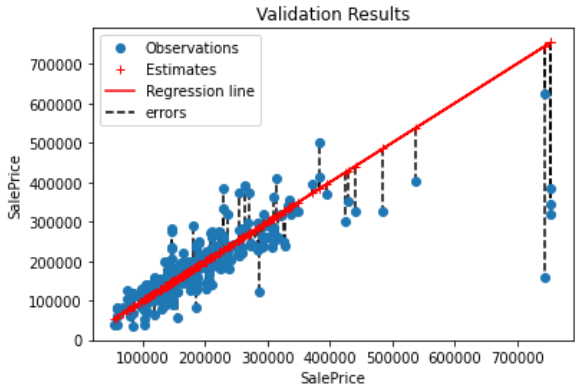




5.4: Gradient Boosted Tree Regression

Our analysis concluded that our estimator showed a high level of correlation and a lower root-mean square error within our training set (r^2 = 98.95%, RMSE = 8337.84), our validation results posed a different picture. Within our validation results, we had low r^2 (38.43%) and higher RMSE (58546.45). This would indicate that this “Gradient boosted tree regression” may not be the most suitable when predicting house prices in Ames, Iowa.





Feature Importance

During our analysis we also made it a priority to determine the attributes that were the most important/influential with regards to impacting home prices in Ames, Iowa. While conducting analysis for our Random Forest Model, our team also derived the importance of the top 10 attributes associated with our dataset (included below).

	index	Importance	attribute
0	3	0.258082	OverallQual
1	42	0.113514	Neighborhood_index
2	24	0.081028	GarageCars
3	52	0.075043	ExterQual_index
4	14	0.067784	GrLivArea
5	10	0.050782	TotalBsmtSF
6	55	0.039779	BsmtQual_index
7	25	0.035347	GarageArea
8	11	0.028692	1stFlrSF
9	64	0.023142	KitchenQual_index

This is one of the most valuable components of our analysis, as it highlights the attributes that people are most concerned about when purchasing a home. Interestingly, the overall quality of the house remains the top priority for those investing in a house. Neighborhood also seems very important, most likely because families want to make sure that it is safe and secure within the general vicinity of the home they are potentially buying. Garage also seems to be an unexpected vital component during a house purchase, indicating that people prioritize how much equipment and cars they can store comfortably in their garage.

6. Conclusion

Due to there being multiple factors and variables that determine how much a home would sell for, we included numerous attributes in our analysis that were determined to have an impact on the house's final sale price. Cleaning the data posed numerous challenges. For example, much of our data is solely structural and temporal in nature which captures just the structural characteristics of the house such as lot size, living area and such. During data preparation it was also imperative that we replace any missing data as for some attributes there was a significant amount missing. While conducting data analysis it was also very important that our data models retained a high level of accuracy by isolating the specific variables that were determined to have tremendous influence on the sale price of homes while dropping the attributes that could skew the data or had a negligible impact.

Our findings would indicate that the Random Forest was the most accurate when analyzing our validation sets due to a lower RMSE and a higher r^2. We did not run on the test set provided on Kaggle as there was no estimate for us to test our predictions against. But on the training set our models worked well. Given the difference in validation results between Random Forest, Linear Regression, GBT & Decision Tree, further examination is warranted to determine if one model does supersede the other in terms of accuracy if we conducted on analysis using different methods. Additionally, a concern for us was also the possibility of overtraining and overfitting our models in some instances, though our team tried to account for this as best as we could. Nevertheless, based on our findings it would seem the Random Forest model is the best predictor.

Finally, one of the most interesting developments from our analysis was the importance of the various attributes in our dataset with regards to predicting the sale price of homes. It was interesting to notice which were the features in a home most valued for a homeowner and how this may impact their purchasing decisions. In conclusion, based on the dataset provided our team was able to create a model that

could successfully predict the sale price of homes in Ames, Iowa based on the attributes from the Kaggle Dataset we used for the purpose of this analysis.